

Chapter 1

Propositional Logic

1.1 Syntax

We begin with a countably infinite set of atomic propositions $\mathcal{P} = \{p_0, p_1, \dots\}$ and two logical connectives $\neg$ (read as *not*) and $\vee$ (read as *or*).

The set Φ of formulas of propositional logic is the smallest set satisfying the following conditions:

- Every atomic proposition p is a member of Φ .
- If α is a member of Φ , so is $(\neg\alpha)$.
- If α and β are members of Φ , so is $(\alpha \vee \beta)$.

We shall normally omit parentheses unless we need to explicitly clarify the structure of a formula. We follow the convention that $\neg$ binds more tightly than $\vee$. For instance, $\neg\alpha \vee \beta$ stands for $((\neg\alpha) \vee \beta)$.

Exercise 1.1.1 Show that Φ is a countably infinite set. $\dashv$

The fact that Φ is the *smallest* set satisfying this inductive definition provides us with the principle of *structural induction*.

Structural induction principle Let S be a set such that:

- Every atomic proposition p is a member of S .
- If α is a member of S , so is $(\neg\alpha)$.

- If α and β are members of S , so is $(\alpha \vee \beta)$.

Then, $\Phi \subseteq S$.

1.2 Semantics

To assign meaning to formulas, we begin by assigning meaning to the atomic propositions. Let $\top$ denote the truth value *true* and $\perp$ the truth value *false*.

- A *valuation* v is a function $v : \mathcal{P} \rightarrow \{\top, \perp\}$.

We can also think of a valuation as a subset of $\mathcal{P}$ —if $v : \mathcal{P} \rightarrow \{\top, \perp\}$, then $v \subseteq \mathcal{P} = \{p \mid v(p) = \top\}$. Thus, the set of all valuations is $2^{\mathcal{P}}$, the set of all subsets of $\mathcal{P}$.

We extend each valuation $v : \mathcal{P} \rightarrow \{\top, \perp\}$ to a map $\hat{v} : \Phi \rightarrow \{\top, \perp\}$ as follows:

- For $p \in \mathcal{P}$, $\hat{v}(p) = v(p)$.
- For α of the form $\neg\beta$, $\hat{v}(\alpha) = \begin{cases} \top & \text{if } \hat{v}(\beta) = \perp \\ \perp & \text{otherwise} \end{cases}$
- For α of the form $\beta \vee \gamma$, $\hat{v}(\alpha) = \begin{cases} \perp & \text{if } \hat{v}(\beta) = \hat{v}(\gamma) = \perp \\ \top & \text{otherwise} \end{cases}$

The principle of structural induction can be used to formally argue that $\hat{v}$ is well-defined (that is, $\hat{v}$ is indeed a function and is defined for all formulas).

Just as v can be defined as a subset of $\mathcal{P}$, $\hat{v}$ can be defined as a subset of Φ —namely, $\hat{v} = \{\alpha \mid \hat{v}(\alpha) = \top\}$.

Exercise 1.2.1 We saw that every subset of $\mathcal{P}$ defines a valuation v . Does every subset of Φ define an extended valuation $\hat{v}$? $\dashv$

Since every valuation v gives rise to a unique extension $\hat{v}$, we shall always denote $\hat{v}$ as just v .

Derived connectives It will be convenient to introduce some additional connectives when discussing propositional logic.

$$\begin{aligned}\alpha \wedge \beta &\stackrel{\text{def}}{=} \neg(\neg\alpha \vee \neg\beta) \\ \alpha \supset \beta &\stackrel{\text{def}}{=} \neg\alpha \vee \beta \\ \alpha \equiv \beta &\stackrel{\text{def}}{=} (\alpha \supset \beta) \wedge (\beta \supset \alpha)\end{aligned}$$

The connective $\wedge$ is read as *and*, $\supset$ as *implies* and $\equiv$ as *if and only if*.

Exercise 1.2.2 Express $v(\alpha \wedge \beta)$, $v(\alpha \supset \beta)$ and $v(\alpha \equiv \beta)$ in terms of $v(\alpha)$ and $v(\beta)$. $\dashv$

Exercise 1.2.3 According to the Pigeonhole Principle, if we try to place $n+1$ pigeons in n pigeonholes, then at least one pigeonhole must have two or more pigeons. For $i \in \{1, 2, \dots, n+1\}$ and $j \in \{1, 2, \dots, n\}$, let the atomic proposition p_{ij} denote that the i^{th} pigeon is placed in the j^{th} pigeonhole. Write down a formula expressing the Pigeonhole Principle. What is the length of your formula as a function of n ? $\dashv$

Satisfiability and validity A formula α is said to be *satisfiable* if there is a valuation v such that $v(\alpha) = \top$. We write $v \models \alpha$ to indicate that $v(\alpha) = \top$.

The formula α is said to be *valid* if $v \models \alpha$ for every valuation v . We write $\models \alpha$ to indicate that α is valid. We also refer to valid formulas of propositional logic as *tautologies*.

Example 1 Let p be an atomic proposition. The formula p is satisfiable. The formula $p \vee \neg p$ is valid. The formula $p \wedge \neg p$ is not satisfiable.

The following observation connects the notions of satisfiability and validity.

Proposition 1.2.4 Let α be a formula. α is valid iff $\neg\alpha$ is not satisfiable.

In applications of logic to computer science, a central concern is to develop algorithms to check for satisfiability and validity of formulas. The preceding remark shows that the two notions are dual: an algorithm which tests validity of formulas can be converted into one for testing satisfiability and vice versa.

In principle, testing the validity of a formula α involves checking its truth value across an uncountable number of valuations. However, it is sufficient to look at the effect of valuations on the atomic propositions mentioned in α .

Let us define $\text{Voc}(\alpha)$, the *vocabulary* of α , as follows:

- For $p \in \mathcal{P}$, $\text{Voc}(p) = \{p\}$.
- If $\alpha = \neg\beta$, then $\text{Voc}(\alpha) = \text{Voc}(\beta)$.
- If $\alpha = \beta \vee \gamma$, then $\text{Voc}(\alpha) = \text{Voc}(\beta) \cup \text{Voc}(\gamma)$.

Proposition 1.2.5 *Let α be a formula and v_1, v_2 be valuations. If v_1 and v_2 agree on $\text{Voc}(\alpha)$ then $v_1(\alpha) = v_2(\alpha)$.*

This justifies the familiar algorithm for testing validity: build a truth-table for the propositions mentioned in α and check if all rows yield the value $\top$.

1.3 Axiomatisations

Though we have a straightforward algorithm for testing validity of formulas in propositional logic, such algorithms do not exist for more complicated logical systems. In particular, there is no such algorithm for first-order logic.

However, it is still possible to effectively enumerate all the valid formulas of first-order logic. One way of presenting such an enumeration is through an axiomatisation of the logic. To prepare the ground for studying axiomatisations of more complex logics, we begin with an axiomatisation for propositional logic.

Axiom System AX The axiom system AX consists of three axioms and one inference rule.

$$\begin{array}{ll}
 (\text{A1}) & \alpha \supset (\beta \supset \alpha) \\
 (\text{A2}) & (\alpha \supset (\beta \supset \gamma)) \supset ((\alpha \supset \beta) \supset (\alpha \supset \gamma)) \\
 (\text{A3}) & (\neg\beta \supset \neg\alpha) \supset ((\neg\beta \supset \alpha) \supset \beta) \\
 (\text{Modus Ponens, or MP}) & \frac{\alpha, \alpha \supset \beta}{\beta}
 \end{array}$$

The rule MP is read as follows—from α and $\alpha \supset \beta$, infer β . It is important to note that these are *axiom schemes*—that is, they are not actual formulas but templates which

can be instantiated into real formulas by consistently substituting concrete formulas for α , β and γ . For instance, if $p, q \in \mathcal{P}$, $p \supset (q \supset p)$ is an instance of axiom (A1). An alternate way to present such an axiomatisation is to list the axioms as concrete formulas and have an additional inference rule to permit uniform substitution of new formulas into an existing formula.

Derivations A *derivation* of α using the axiom system AX is a finite sequence of formulas $\beta_1, \beta_2, \dots, \beta_n$ such that:

- $\beta_n = \alpha$
- For each $i \in \{1, 2, \dots, n\}$, β_i is either an instance of one of the axioms (A1)–(A3), or is obtained by applying the rule (MP) to formulas β_j, β_k , where $j, k < i$ —that is, β_k is of the form $\beta_j \supset \beta_i$.

We write $\vdash_{AX} \alpha$ to denote that α is derivable using the axiom system AX and say that α is a *thesis* of the system. We will normally omit the subscript AX .

Here is an example of a derivation using our axiom system.

- | | |
|--|--------------------|
| 1. $(p \supset ((p \supset p) \supset p)) \supset ((p \supset (p \supset p)) \supset (p \supset p))$ | Instance of (A2) |
| 2. $p \supset ((p \supset p) \supset p)$ | Instance of (A1) |
| 3. $(p \supset (p \supset p)) \supset (p \supset p)$ | From 1 and 2 by MP |
| 4. $p \supset (p \supset p)$ | Instance of (A1) |
| 5. $p \supset p$ | From 3 and 4 by MP |

Exercise 1.3.1 Show that $(\neg\beta \supset \neg\alpha) \equiv (\alpha \supset \beta)$ is a thesis of AX . $\dashv$

The axiom system we have presented is called a Hilbert-style axiomatisation. There are several other ways of presenting axiomatisations. One common alternative to Hilbert-style systems is the *sequent calculus* notation due to Gentzen. Typically, Hilbert-style axiomatisations have a large number of axioms and very few inference rules, while sequent calculi have very few axioms and a large number of inference rules. Sequent calculi are often easier to work with when searching for derivations, but are also more complicated from a technical point of view. We shall look at sequent calculi later, when we come to first-order logic.

Another fact worth remembering is that the axiom system AX defined here is just one of many possible Hilbert-style axiom systems for propositional logic.

The main technical result we would like to establish is that the set of formulas derivable using AX is precisely the set of valid formulas of propositional logic.

Theorem 1.3.2 *For all formulas α , $\vdash \alpha$ iff $\models \alpha$.*

We break up the proof of this theorem into two parts. The first half is to show that every thesis of AX is valid. This establishes the *soundness* of the axiom system,

Lemma 1.3.3 (Soundness) *For all formulas α , if $\vdash \alpha$ then $\models \alpha$.*

PROOF If $\vdash \alpha$, then we can exhibit a derivation $\beta_1, \beta_2, \dots, \beta_n$ of α . Formally, the proof of the lemma is by induction on the length of this derivation. Since every formula in the sequence $\beta_1, \beta_2, \dots, \beta_n$ is either an instance of one of the axioms or is obtained by applying the rule (MP), it suffices to show that all the axioms define valid formulas and that (MP) preserves validity—in other words, if α is valid and $\alpha \supset \beta$ is valid, then β is valid. This is straightforward and we omit the details. $\dashv$

The other half of Theorem 1.3.2 is more difficult to establish. We have to argue that every valid formula is derivable. Formally, this would show that our axiomatisation is *complete*.

We follow the approach of the logician Leon Henkin and attack the problem indirectly. Consider the contrapositive of the statement we want to prove—that is, if a formula α is *not* a thesis, then it is *not* valid.

Consistency We write $\nvdash \alpha$ to denote that α is not a thesis. We say that α is *consistent* (with respect to AX) if $\nvdash \neg \alpha$.

Exercise 1.3.4

- (i) Show that $\alpha \vee \beta$ is consistent iff either α is consistent or β is consistent.
- (ii) Show that if $\alpha \wedge \beta$ is consistent then both α and β are consistent. Is the converse true?
- (iii) Suppose that $\vdash \alpha \supset \beta$. Which of the following is true?
 - (a) If α is consistent then β is consistent.
 - (b) If β is consistent then α is consistent. $\dashv$

By Proposition 1.2.4 we know that α is not valid iff $\neg \alpha$ is satisfiable. Suppose we can show the following.

Lemma 1.3.5 (Henkin) *For all formulas β , if β is consistent then β is satisfiable.*

We can then argue that our axiomatisation is complete. Consider a formula β which is not derivable. It can be shown that $\neg\neg\beta \supset \beta$ is a thesis. If β is not derivable, neither is $\neg\neg\beta$ —otherwise, we can use the rule MP to derive β from $\neg\neg\beta \supset \beta$. Since $\not\vdash \neg(\neg\beta)$, $\neg\beta$ is consistent. By Lemma 1.3.5, $\neg\beta$ is satisfiable. Hence, by Proposition 1.2.4, β is not valid.

1.4 Maximal Consistent Sets and Completeness

To prove Lemma 1.3.5, we extend the notion of consistency from a single formula to sets of formulas. A finite set of formulas $X = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is consistent if the formula $\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n$ is consistent—that is, $\not\vdash \neg(\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n)$. An arbitrary set of formulas $X \subseteq \Phi$ is consistent if every finite subset of X is consistent. (Henceforth, $Y \subseteq_{\text{fin}} X$ denotes that Y is a finite subset of X .)

A *maximal consistent set (MCS)* is a consistent set which cannot be extended by adding any formulas. In other words, $X \subseteq \Phi$ is an MCS iff X is consistent and for each formula $\alpha \notin X$, $X \cup \{\alpha\}$ is inconsistent.

Lemma 1.4.1 (Lindenbaum) *Every consistent set can be extended to an MCS.*

PROOF Let X be an arbitrary consistent set. Let $\alpha_0, \alpha_1, \alpha_2, \dots$ be an enumeration of Φ .

We define an infinite sequence of sets $X_0, X_1, X_2, \dots$ as follows.

- $X_0 = X$
- For $i \geq 0$, $X_{i+1} = \begin{cases} X_i \cup \{\alpha_i\} & \text{if } X_i \cup \{\alpha_i\} \text{ is consistent} \\ X_i & \text{otherwise} \end{cases}$

Each set in this sequence is consistent, by construction, and $X_0 \subseteq X_1 \subseteq X_2 \subseteq \dots$. Let $Y = \bigcup_{i \geq 0} X_i$. We claim that Y is an MCS extending X . To establish this, we have to show that Y is consistent and that it is maximal.

If Y is not consistent, then there is a subset $Z \subseteq_{\text{fin}} Y$ which is inconsistent. Let $Z = \{\beta_1, \beta_2, \dots, \beta_n\}$. We can write Z as $\{\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_n}\}$ where the indices correspond to our enumeration of Φ . Then it is clear that $Z \subseteq_{\text{fin}} X_{j+1}$ in the sequence $X_0 \subseteq X_1 \subseteq X_2 \subseteq \dots \subseteq Y$, where $j = \max(i_1, i_2, \dots, i_n)$. This implies that X_{j+1} is inconsistent, which is a contradiction.

Having established that Y is consistent, we show that it is maximal. Suppose that $Y \cup \{\beta\}$ is consistent for some formula $\beta \notin Y$. Let $\beta = \alpha_j$ in our enumeration of Φ . Since $\alpha_j \notin Y$, α_j was not added at step $j+1$ in our construction. This means that $X_j \cup \{\alpha_j\}$ is inconsistent. In other words, there exists $Z \subseteq_{\text{fin}} X_j$ such that $Z \cup \{\alpha_j\}$ is inconsistent. Since $X_j \subseteq Y$, we must have $Z \subseteq_{\text{fin}} Y$ as well, which contradicts the assumption that $Y \cup \{\alpha_j\}$ is consistent. $\neg$

Maximal consistent sets have a rich structure which we shall exploit to prove completeness.

Lemma 1.4.2 *Let X be a maximal consistent set. Then:*

- (i) *For all formulas α , $\alpha \in X$ iff $\neg\alpha \notin X$.*
- (ii) *For all formulas α, β , $\alpha \vee \beta \in X$ iff $\alpha \in X$ or $\beta \in X$.*

We postpone the proof of these properties and first show how they lead to completeness.

Maximal consistent sets and valuations Let X be an MCS. Define the valuation v_X to be the set $\{p \in \mathcal{P} \mid p \in X\}$ —in other words, $v_X(p) = \top$ iff $p \in X$.

Proposition 1.4.3 *Let X be an MCS. For all formulas α , $v_X \models \alpha$ iff $\alpha \in X$.*

PROOF The proof is by induction on the structure of α .

Basis: $\alpha = p$, where $p \in \mathcal{P}$. Then, $v_X \models p$ iff (by the definition of v_X) $p \in X$.

Induction step: There are two cases to consider—when α is of the form $\neg\beta$ and when α is of the form $\beta \vee \gamma$.

$(\alpha = \neg\beta)$ $v_X \models \neg\beta$ iff (by the definition of valuations) $v_X \not\models \beta$ iff (by the induction hypothesis) $\beta \notin X$ iff (by the properties satisfied by MCSs) $\neg\beta \in X$.

$(\alpha = \beta \vee \gamma)$ $v_X \models \beta \vee \gamma$ iff (by the definition of valuations) $v_X \models \beta$ or $v_X \models \gamma$ iff (by the induction hypothesis) $\beta \in X$ or $\gamma \in X$ iff (by the properties satisfied by MCSs) $\beta \vee \gamma \in X$. $\neg$

Thus, every MCS X defines a canonical valuation v_X which satisfies precisely those formulas that belong to X . (Conversely, every valuation also defines an MCS in a canonical way: given a valuation v , $X_v = \{\alpha \mid v \models \alpha\}$. It is not difficult to establish that the valuation v_{X_v} generated by X_v is exactly the same as v .)

Proposition 1.4.3 immediately yields a proof of Henkin's lemma.

Proof of Lemma 1.3.5: Let α be a consistent formula. By Lindenbaum's Lemma, $\{\alpha\}$ can be extended to an MCS X . By Proposition 1.4.3, $v_X \models \alpha$ since $\alpha \in X$. Thus, α is satisfiable. $\dashv$

To complete our argument, we have to prove Lemma 1.4.2.

Proof Sketch of Lemma 1.4.2: Let X be an MCS.

- (i) For every formula α , we have to show that $\alpha \in X$ iff $\neg\alpha \notin X$.

We first show that $\{\alpha, \neg\alpha\} \not\subseteq X$. For this, we need the fact that $\alpha \supset \neg\neg\alpha$ and $\neg\neg\alpha \supset \alpha$ are both derivable using AX. We omit these derivations.

We know that $\alpha \supset \alpha$, or, equivalently, $\neg\alpha \vee \alpha$ is a thesis. From this, we can derive $\neg\neg(\neg\alpha \vee \alpha)$. But $\neg(\neg\alpha \vee \alpha)$ is just $\alpha \wedge \neg\alpha$, so we have $\neg(\alpha \wedge \neg\alpha)$ as a thesis. This means that $\{\alpha, \neg\alpha\}$ is inconsistent, whence it cannot be a subset of X (recalling that X is consistent).

Next we show that at least one of α and $\neg\alpha$ is in X . Suppose neither formula belongs to X . Since X is an MCS, there must be sets $B \subseteq_{\text{fin}} X$ and $C \subseteq_{\text{fin}} X$ such that $B \cup \{\alpha\}$ is inconsistent and $C \cup \{\neg\alpha\}$ are inconsistent. Let $B = \{\beta_1, \beta_2, \dots, \beta_n\}$ and $C = \{\gamma_1, \gamma_2, \dots, \gamma_m\}$. Let $\hat{\beta}$ abbreviate the formula $\beta_1 \wedge \beta_2 \wedge \dots \wedge \beta_n$ and $\hat{\gamma}$ abbreviate the formula $\gamma_1 \wedge \gamma_2 \wedge \dots \wedge \gamma_m$. Then, we have $\vdash \neg(\alpha \wedge \hat{\beta})$ and $\vdash \neg(\neg\alpha \wedge \hat{\gamma})$. Rewriting $\wedge$ in terms of $\vee$, this is equivalent to $\vdash \neg\alpha \vee \neg\hat{\beta}$ and $\vdash \neg\neg\alpha \vee \neg\hat{\gamma}$. From this, we can conclude that $\vdash \alpha \supset \neg\hat{\beta}$ and $\vdash \neg\alpha \supset \neg\hat{\gamma}$.

We now use that fact that $(\alpha \supset \beta) \supset ((\delta \supset \gamma) \supset ((\alpha \vee \delta) \supset (\beta \vee \gamma)))$ is a thesis. (Once again, we omit the derivation). Instantiating this with $\alpha = \alpha$, $\delta = \neg\alpha$, $\beta = \neg\hat{\beta}$ and $\gamma = \neg\hat{\gamma}$ we can derive $(\alpha \vee \neg\alpha) \supset (\neg\hat{\beta} \vee \neg\hat{\gamma})$. Since $\vdash \alpha \vee \neg\alpha$, we get $\vdash \neg\hat{\beta} \vee \neg\hat{\gamma}$. By rewriting $\vee$ in terms of $\wedge$, we can derive $\neg(\hat{\beta} \wedge \hat{\gamma})$. But this implies that $(B \cup C) \subseteq_{\text{fin}} X$ is inconsistent, which is a contradiction.

- (ii) The proof of the second part follows in a similar manner, assuming the derivability of appropriate formulas. We omit the details. $\dashv$

1.5 Compactness and Strong Completeness

Often, we are not interested in absolute validity, but in restricted validity. Rather than asking whether a formula α is *always* true, we ask whether α is true in all valuations

which satisfy certain properties. One way of restricting the class of valuations under consideration is to specify a set of formulas X and only look at those valuations where X is true. If α is true wherever the formulas from X are true, then α is a logical consequence of X .

Logical consequence Let X be a set of formulas and v a valuation. We write $v \models X$ to denote that $v \models \beta$ for every formula $\beta \in X$. A formula α is a *logical consequence* of X , written $X \models \alpha$, if for every valuation v such that $v \models X$ it is also the case that $v \models \alpha$.

The notion of logical consequence is central to the way we formalise mathematics. For instance, when we study algebraic structures such as groups, we first formulate axioms which characterise groups. Any theorem we prove about groups can be rephrased as a statement which is a logical consequence of these axioms: in other words, the theorem is true whenever the group axioms are also true.

As with validity, we now look at a syntactic approach to logical consequence.

Derivability Let X be a set of formulas. We say that a formula α is *derivable from* X , written $X \vdash \alpha$ if there exists a sequence $\alpha_1, \alpha_2, \dots, \alpha_n$ of formulas such that $\alpha_n = \alpha$ and for $i \in \{1, 2, \dots, n\}$, α_i is either a member of X , or an instance of one of the axioms (A1)–(A3) of AX , or is derived from α_j, α_k , $j, k < i$, using the inference rule MP. (Notice that unlike axioms, we cannot use the formulas in X as templates to generate new formulas for use in a derivation. The formulas in X are concrete formulas and must be used “as is”.)

The theorem we would like to prove is the following.

Theorem 1.5.1 (Strong Completeness) *Let $X \subseteq \Phi$ and $\alpha \in \Phi$. Then, $X \models \alpha$ iff $X \vdash \alpha$.*

It is possible to prove this directly using a technique similar to the one used to prove the soundness and completeness of AX (see Exercise 1.5.7). However, we will prove it indirectly using two auxiliary results which are of independent interest—the Deduction Theorem and the Compactness Theorem.

We begin with the Deduction Theorem, which is a statement about derivability.

Theorem 1.5.2 (Deduction) *Let $X \subseteq \Phi$ and $\alpha, \beta \in \Phi$. Then, $X \cup \{\alpha\} \vdash \beta$ iff $X \vdash \alpha \supset \beta$.*

PROOF ($\Leftarrow$) Suppose that $X \vdash \alpha \supset \beta$. Then, by the definition of derivability, $X \cup \{\alpha\} \vdash \alpha \supset \beta$ as well. Since $\alpha \in X \cup \{\alpha\}$, $X \cup \{\alpha\} \vdash \alpha$. Applying MP, we get $X \cup \{\alpha\} \vdash \beta$.

($\Rightarrow$) Suppose that $X \cup \{\alpha\} \vdash \beta$. Then, there is a derivation $\beta_1, \beta_2, \dots, \beta_n$ of β . The proof is by induction on n .

If $n = 1$, then β is either an instance of an axiom or a member of $X \cup \{\alpha\}$. If β is an instance of an axiom, then $X \vdash \beta$ as well. Further, from axiom (A1), $X \vdash \beta \supset (\alpha \supset \beta)$. Applying MP, we get $X \vdash \alpha \supset \beta$.

If $\beta \in X$, there are two cases to consider. If $\beta \in X \setminus \{\alpha\}$, then $X \vdash \beta$. Once again we have $X \vdash \beta \supset (\alpha \supset \beta)$ and hence $X \vdash \alpha \supset \beta$. On the other hand, if $\beta = \alpha$, we have $X \vdash \alpha \supset \alpha$ from the fact that $\alpha \supset \alpha$ is derivable in AX.

If $n > 1$, we look the justification for adding $\beta_n = \beta$ to the derivation. If β_n is an instance of an axiom or a member of $X \cup \{\alpha\}$, we can use the same argument as in the base case to show $X \vdash \alpha \supset \beta$.

On the other hand, if β_n was derived using MP, there exist β_i and β_j , with $i, j < n$ such that β_j is of the form $\beta_i \supset \beta_n$. By axiom (A2), $X \vdash (\alpha \supset (\beta_i \supset \beta_n)) \supset ((\alpha \supset \beta_i) \supset (\alpha \supset \beta_n))$. By the induction hypothesis, we know that $X \vdash \alpha \supset (\beta_i \supset \beta_n)$ and $X \vdash \alpha \supset \beta_i$. Applying MP twice, we get $X \vdash \alpha \supset \beta_n$. $\dashv$

The Deduction Theorem reflects a method of proof which is common in mathematics—proving that property x implies property y is equivalent to assuming x and inferring y .

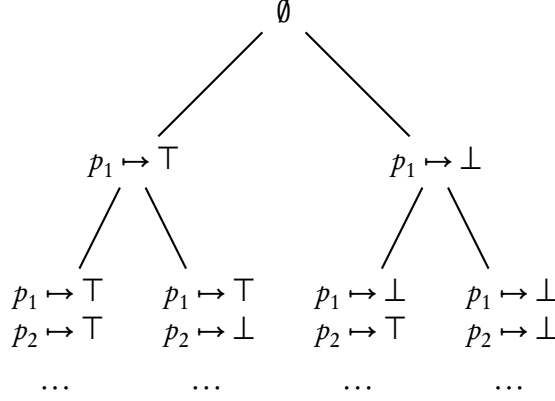
The second step in proving Strong Completeness is the Compactness Theorem, which is a statement about logical consequence. To prove this we need the following lemma about trees, due to König.

Lemma 1.5.3 (König) *Let T be a finitely branching tree—that is, every node has a finite number of children (though this number may be unbounded). If T has infinitely many nodes, then T has an infinite path.*

PROOF Let T be a finitely branching tree with infinitely many nodes. Call a node x in T *bad* if the subtree rooted at x has infinitely many nodes. Clearly, if a node x is bad, at least one of its children must be bad: x has only finitely many children and if all of them were good, the subtree rooted at x would be finite.

We now construct an infinite path $x_0 x_1 x_2 \dots$ in T . Since T has an infinite number of nodes, the root of T is a bad node. Let x_0 be the root of T . It has at least one bad successor. Pick one of the bad successors of x_0 and designate it x_1 . Pick one of the bad successors of x_1 and designate it x_2 , and so on. $\dashv$

Theorem 1.5.4 (Compactness) *Let $X \subseteq \Phi$ and $\alpha \in \Phi$. Then $X \models \alpha$ iff there exists $Y \subseteq_{fn} X$, $Y \models \alpha$.*

Figure 1.1: The tree T in the proof of Lemma 1.5.5

We shall first prove the following related result. Let X be a set of formulas. We say that X is satisfiable if there exists a valuation v such that $v \models X$.

Lemma 1.5.5 (Finite satisfiability) *Let $X \subseteq \Phi$. Then, X is satisfiable iff every $Y \subseteq_{\text{fin}} X$ is satisfiable.*

PROOF ($\Rightarrow$) Suppose X is satisfiable. Then, there is a valuation v such that $v \models X$. Clearly, $v \models Y$ for each $Y \subseteq_{\text{fin}} X$ as well.

($\Leftarrow$) Suppose X is *not* satisfiable. We have to show that there exists $Y \subseteq_{\text{fin}} X$ which is not satisfiable.

Assume that our set of atomic propositions $\mathcal{P}$ is enumerated $\{p_1, p_2, \dots\}$. Let $\mathcal{P}_0 = \emptyset$ and for $i \in \{1, 2, \dots\}$, let $\mathcal{P}_i = \{p_1, p_2, \dots, p_i\}$. For $i \in \{1, 2, \dots\}$, let Φ_i be the set of formulas generated using only atomic propositions from $\mathcal{P}_i$ and let $X_i = X \cap \Phi_i$.

We construct a tree T whose nodes are valuations over the sets $\mathcal{P}_i$, $i \in \{0, 1, 2, \dots\}$. More formally, the set of nodes is given by $\{v \mid \exists i \in \{0, 1, 2, \dots\}. v : \mathcal{P}_i \rightarrow \{\top, \perp\}\}$. The root of T is the unique function $\emptyset \rightarrow \{\top, \perp\}$.

The relation between nodes is given as follows. Let $v : \mathcal{P}_i \rightarrow \{\top, \perp\}$. Then v has two children $v', v'' : \mathcal{P}_{i+1} \rightarrow \{\top, \perp\}$, where v' extends v to $\mathcal{P}_{i+1}$ by setting p_{i+1} to $\top$ and v'' extends v to $\mathcal{P}_{i+1}$ by setting p_{i+1} to $\perp$. More formally, for each $p \in \mathcal{P}_i$, $v'(p) = v''(p) = v(p)$ and $v'(p_{i+1}) = \top$ and $v''(p_{i+1}) = \perp$. (See Figure 1.1).

Observe that T is a complete infinite binary tree. The nodes at level i of the tree consist of all possible valuations over $\mathcal{P}_i$ —there are precisely 2^i such valuations for

each i . Notice that if v_j at level j is an ancestor of v_i at level i then v_i agrees with v_j on the atomic propositions in $\mathcal{P}_j$.

The infinite paths in T are in 1-1 correspondence with valuations over $\mathcal{P}$. Let $\pi = v_0 v_1 v_2 \dots$ be an infinite path in the tree. The valuation $v_\pi : \mathcal{P} \rightarrow \{\top, \perp\}$ is given by $p_i \mapsto v_i(p_i)$ for $i \in \{1, 2, \dots\}$. Conversely, given a valuation $v : \mathcal{P} \rightarrow \{\top, \perp\}$, we can define a unique path $\pi_v = v_0 v_1 v_2 \dots$ by setting v_0 to be the root of T and $v_i : \mathcal{P}_i \rightarrow \{\top, \perp\}$ to be the restriction of v to $\mathcal{P}_i$ —that is, for all $p \in \mathcal{P}_i$, $v_i(p) = v(p)$. It is easy to verify that these two maps are inverses of each other.

Let us call a node v in T bad if $v(\beta) = \perp$ for some $\beta \in X$. Clearly, if v is bad, then so is every valuation in the subtree rooted at v . We prune T by deleting all bad nodes which also have bad ancestors. (Equivalently, along any path in T , we retain only those nodes upto and including the first bad node along the path.) It is not difficult to verify that the set of nodes which remains forms a subtree T' of T all of whose leaf nodes are bad and all of whose non-leaf nodes are not bad.

We claim that T' has only a finite number of nodes. Assuming that this is true, let the set of leaf nodes of T' be $\{v_1, v_2, \dots, v_m\}$. Since each v_i is bad, there is a corresponding formula $\beta_i \in X$ such that $v_i(\beta_i) = \perp$. We claim that $\{\beta_1, \beta_2, \dots, \beta_m\} \subseteq_{\text{fin}} X$ is not satisfiable. Consider any valuation v . The corresponding path π_v must pass through one of the nodes in $\{v_1, v_2, \dots, v_m\}$, say v_j . But then, $v_{\pi_v}(\beta_j) = v_j(\beta_j) = \perp$. Thus, $v \not\models \{\beta_1, \beta_2, \dots, \beta_m\}$.

To see why T' must be finite, suppose instead that it has an infinite set of nodes. Then, by König's Lemma, it contains an infinite path $\pi = v_0 v_1 v_2 \dots$ such that none of the nodes along this path is bad. The path π is also an infinite path in T . We know that π defines a valuation v_π . Consider any formula $\beta \in X$. Then $\beta \in X_j$ for some $j \in \{1, 2, \dots\}$, so $v_\pi(\beta) = v_j(\beta) = \top$. Thus, $v_\pi \models X$, which contradicts our assumption that X is not satisfiable. $\dashv$

We can now complete our proof of compactness.

Proof of Theorem 1.5.4 (Compactness):

($\Leftarrow$) If $Y \subseteq_{\text{fin}} X$ and $Y \models \alpha$ then it is clear that $X \models \alpha$. For, if $v \models X$, then $v \models Y$ as well and, by the assumption that $Y \models \alpha$, $v \models \alpha$ as required.

($\Rightarrow$) For all $Z \subseteq \Phi$ and all $\beta \in \Phi$, it is clear that $Z \models \beta$ iff $Z \cup \{\neg\beta\}$ is not satisfiable.

Suppose $X \models \alpha$. Then, $X \cup \{\neg\alpha\}$ is not satisfiable. By Lemma 1.5.5, there is a subset $Y \subseteq_{\text{fin}} X \cup \{\neg\alpha\}$ such that Y is not satisfiable. Thus, $(Y \setminus \{\neg\alpha\}) \cup \{\neg\alpha\}$ is not satisfiable either, where $(Y \setminus \{\neg\alpha\}) \subseteq_{\text{fin}} X$. This implies that $Y \setminus \{\neg\alpha\} \models \alpha$. $\dashv$

With the Deduction Theorem and the Compactness Theorem behind us, we can prove Strong Completeness.

Proof of Theorem 1.5.1 (Strong Completeness):

To show that $X \vdash \alpha$ implies $X \models \alpha$ is routine. Conversely, suppose that $X \models \alpha$. By compactness, there is a finite subset $Y \subseteq_{\text{fin}} X$ such that $Y \models \alpha$. Let $Y = \{\beta_1, \beta_2, \dots, \beta_m\}$. It is then easy to see that $\beta_1(\supset (\beta_2(\supset \dots (\beta_m \supset \alpha) \dots))$ is valid. Hence, by the completeness theorem for propositional logic, $\vdash \beta_1(\supset (\beta_2(\supset \dots (\beta_m \supset \alpha) \dots))$. Applying the Deduction Theorem m times we get $\{\beta_1, \beta_2, \dots, \beta_m\} \vdash \alpha$. Since $\{\beta_1, \beta_2, \dots, \beta_m\} \subseteq X$, it follows that $X \vdash \alpha$. $\dashv$

Observe that we could alternatively derive compactness from strong completeness. If $X \models \alpha$ then, by strong completeness, $X \vdash \alpha$. We let $Y \subseteq_{\text{fin}} X$ be the subset of formulas actually used in the derivation of α . Thus, $Y \vdash \alpha$ as well. By the other half of strong completeness, $Y \models \alpha$.

We conclude our discussion of propositional logic with two exercises. The first leads to an alternative proof of compactness which is more along the lines of the completeness proof for propositional logic. The second exercise leads to a direct proof of strong completeness.

Exercise 1.5.6 (Compactness)

Let X be a set of formulas. X is said to be a *finitely satisfiable set (FSS)* if every $Y \subseteq_{\text{fin}} X$ is satisfiable.

Equivalently, X is an FSS if there is no finite subset $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of X such that $\neg(\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n)$ is valid.

(Note that if X is an FSS we are not promised a single valuation v which satisfies every finite subset of X . Each finite subset could be satisfied by a *different* valuation).

Show that:

- (i) Every FSS can be extended to a maximal FSS.
- (ii) If X is a maximal FSS then:
 - (a) For every formula α , $\alpha \in X$ iff $\neg\alpha \notin X$.
 - (b) For all formulas α, β , $(\alpha \vee \beta) \in X$ iff $(\alpha \in X \text{ or } \beta \in X)$.
- (iii) Every maximal FSS X generates a valuation v_X such that for every formula α , $v_X \models \alpha$ iff $\alpha \in X$.

From these facts conclude that:

- (iv) Any FSS X is simultaneously satisfiable (that is, for any FSS X , there exists v_X such that $v_X \models X$).

- (v) For all X and all α , $X \models \alpha$ iff there exists $Y \subseteq_{\text{fin}} X$ such that $Y \models \alpha$. $\dashv$

Exercise 1.5.7 (Strong Completeness)

We define a new notion of consistency. A set X is said to be *consistent* if there is no formula α such that $X \vdash \alpha$ and $X \vdash \neg\alpha$.

Show that:

- (i) X is consistent iff every finite subset of X is consistent.
- (ii) Every consistent set X can be extended to a maximal consistent set (MCS).
- (iii) Every MCS X generates a valuation v_X such that for all formulas α , $v_X \models \alpha$ iff $\alpha \in X$.
- (iv) Every consistent set X is satisfiable: that is, there exists a valuation v_X such that $v_X \models X$.
- (v) If $X \models \alpha$ then $X \cup \{\neg\alpha\}$ is not consistent.
- (vi) Use the Deduction Theorem to show that that if $X \models \alpha$ then $X \vdash \neg\alpha \supset (\beta \wedge \neg\beta)$ for some formula β .

Conclude that if $X \models \alpha$ then $X \vdash \alpha$. $\dashv$